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### Ropeless fishing system and method

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#### Abstract

Disclosed herein are systems and methods for ropeless fishing. The ropeless fishing system includes a surface subsystem installed on a vessel and at least one submerged subsystem submerged in a body of water. The surface subsystem includes acoustic transceiver and acoustic transducer units, which are configured to generate and omnidirectionally transmit an acoustic interrogation signal into the body of water surrounding the vessel and to directionally receive acoustic reply signals. The submerged subsystems include a compressed gas source and an enclosed flotation bag in fluid communication and an underwater acoustic actuator-transponder unit configured to receive the acoustic interrogation signal transmitted by the surface subsystem and to transmit an acoustic reply signal into the body of water. In addition, the submerged subsystem is further configured to control the flow of gas between the compressed gas source and the enclosed flotation bag to facilitate recovery of the submerged subsystem.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 62/934,566, filed Nov. 13, 2019, the disclosure of which is herein incorporated by reference.

### BACKGROUND

#### 1. Field of the Invention

(1) This invention describes systems and methods to allow the ability to conduct certain fishing operations without the need for vertical lines and buoys attached to fixed fishing gear deployed below the sea surface or onto on the sea bottom.

#### 2. Discussion of Background Information

(2) Commercial fishing methods can be broadly classified as either “fixed” or “mobile.” One class of fixed gear refers to the use of traps or cages to harvest certain species of crustaceans (lobsters, crabs) or shellfish (oysters, mussels). Other types of fixed gear include gill nets to entrap finfish. Mobile gear refers to the use of apparatus towed from moving vessels (bottom trawls, mid-water trawls, long line, clam and scallop dredges, etc.).

(3) This invention focuses on fixed gear which traditionally employs vertical lines attached to surface floats to mark the location of the subsurface gear and allow for its retrieval. These vertical lines present a lethal hazard to marine animals, with particular devastation imparted to the critically endangered North Atlantic Right Whale (NARW), with rope entanglements having been attributed as the cause of death in 85% of all NARW fatalities.

(4) Existing attempts to remove the traditional vertical lines attached to surface floats suffer from an inability to identify the fixed gear once it is deployed below the sea surface. For example, unmarked fixed gear is susceptible to being towed through by mobile gear fisheries (bottom trawl, scallop dredge, etc.) and set over by other fixed gear.

(5) Ropeless fishing systems or buoyless fishing systems, referred to herein collectively as “ropeless fishing systems,” take away the end line and buoy. However, in order for a ropeless fishing system to be practical, the functions of these components must be replaced. For example, in order to fully replace the existing vertical lines attached to surface floats, ropeless fishing systems should provide address the current role of surface floats by providing a means to identify the ownership of submerged fishing gear and marking the location of submerged fishing gear in order to prevent placement conflicts. In addition, ropeless fishing systems should also facilitate the retrieval of the submerged gear by its owner once the fishing gear is deployed on the seafloor.

(6) One approach to ropeless fishing systems involves the use of float release systems such as those disclosed by Lloyd et al. in U.S. Pat. No. 7,534,152 (“Lloyd”) and Abulrassoul et al. in US Publication No. 20130109259 (“Abulrassoul”). While such systems achieve the goal of avoiding lines with surface floats while the fishing gear is positioned on the seafloor, these systems fail to provide a means for the user to identify the location of the fishing gear once it is deployed.

Accordingly, if the gear shifts due to tidal currents or is dragged by mobile fishing gear, there is no way for the owner to identify the new location of the fishing gear. In addition, these systems lack the ability for non-owners to ascertain the existence of fishing gear on the seafloor, increasing the likelihood that non-owners will tow-through or set over such systems.

(7) Another example of an existing attempt at producing a ropeless fishing system is disclosed by Greenberg et al. in US Publication No. 20190110452 (“Greenberg”). Greenberg discloses a ropeless fishing system that uses acoustic sound waves to communicate with submerged fishing gear to enable lifting submerged fishing gear from the seafloor. However, Greenberg fails to

address other significant features of traditional vertical-line-attached-to-surface-float systems such as providing a means to identify the location and the ownership of submerged fishing gear.

(8) These existing technologies fail to adequately address the issues surrounding the requirements for ropeless fishing systems and methods to replace the traditional vertical line and surface float. For example, existing technologies lack the ability to accurately locate and plot the position of submerged fishing gear, do not provide the ability for fishermen to communicate with only their own submerged fishing gear, and do not provide regulators and law enforcement with the ability to monitor the use of submerged fishing gear.

(9) While other devices and methods have been proposed for achieving the goal of ropeless fishing, none of these inventions, taken either singly or in combination, adequately address or resolve the aforementioned problems. Therefore, a need exists for an efficient system and method for ropeless fishing that addresses the need for direct communication, retrieval, underwater locating, and identification of submerged fishing gear, and allows for the results of the underwater positioning to be displayed on almost any modern, commercial chart plotter.

#### SUMMARY OF THE INVENTION

(10) The present invention solves the problems associated with ropeless fishing and provides a reliable and efficient ropeless fishing system and method.

(11) The present invention is directed to a ropeless fishing system that enables a single surface subsystem to communicate with multiple pieces of submerged fishing gear. The ropeless fishing system comprises a surface subsystem installed on a vessel, the surface subsystem comprising an acoustic transceiver unit in communication with an acoustic transducer unit, the surface subsystem configured to generate and omnidirectionally transmit an acoustic interrogation signal into a body of water surrounding the vessel and to directionally receive acoustic reply signals; and at least one submerged subsystem submerged in the body of water, each of the at least one submerged subsystem comprising a compressed gas source and an enclosed flotation bag in fluid communication and an underwater acoustic actuator-transponder unit configured to receive the acoustic interrogation signal transmitted by the surface subsystem and to transmit an acoustic reply signal into the body of water and further configured to control the flow of gas between the compressed gas source and the enclosed flotation bag. The acoustic transducer unit of the ropeless fishing system may further comprise a first transducer configured to directionally receive the acoustic reply signal and a second transducer configured to omnidirectionally transmit the acoustic interrogation signal. In addition, the surface subsystem may be further configured to process each acoustic reply signal generated by each of the at least one submerged subsystem and received by the surface subsystem to determine the location, ownership, and association of each of the at least one submerged subsystem and to display this location, ownership and association information to a user on a human machine interface.

(12) The present invention is also directed to a method of ropeless fishing. The method comprises a first step of providing a ropeless fishing system comprising a surface subsystem installed on a vessel, the surface subsystem comprising an acoustic transceiver unit in communication with an acoustic transducer unit, the surface subsystem configured to omnidirectionally transmit an acoustic interrogation signal into a body of water surrounding the vessel and to directionally receive acoustic reply signals; and at least one submerged subsystem submerged in the body of water, each of the at least one submerged subsystem comprising a compressed gas source and an enclosed flotation bag in fluid communication and an underwater acoustic actuator-transponder unit configured to receive the acoustic signal transmitted by the surface subsystem and to transmit an acoustic signal into the body of water and further configured to control the flow of gas between the compressed gas source and the enclosed flotation bag. The surface subsystem generates and omnidirectionally transmits an acoustic interrogation signal, which is received by one of the at least one submerged subsystem. Once received, the acoustic interrogation signal is processed with the underwater acoustic actuator-transponder unit, which generates and transmits an acoustic reply

signal. The acoustic reply signal is received with the surface subsystem, where the relative bearing of the acoustic reply signal is measured, and slant range and true bearing are calculated. The surface subsystem then calculates the location of the submerged subsystem. Where a plurality of submerged subsystems are deployed in the body of water, the steps of receiving and processing the acoustic reply signal and calculating the subsystem location are repeated for each acoustic reply signal received by the surface subsystem.

(13) The method may further comprise generating and transmitting a second acoustic interrogation signal designed to cause one of the at least one submerged subsystem to rise to the surface of the body of water; receiving the second acoustic interrogation signal with the one of the at least one submerged subsystem; and processing the received acoustic interrogation signal with the underwater acoustic actuator-transponder unit and generating and transmitting a control signal that actuates a solenoid valve, causing gas from the compressed gas source to flow into and inflate the enclosed floatation bag and causing the one of the at least one submerged subsystem to rise to the surface of the body of water.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description, appended claims, and accompanying drawings where:

(2) FIG. 1 depicts an embodiment of the system showing the component subsystems.

(3) FIG. 2 depicts an embodiment of a component subsystem of the system.

(4) FIG. 3 depicts an electronic flow diagram of an embodiment of a component subsystem of the system.

(5) FIG. 4A depicts an embodiment of an acoustic transducer component of the system.

(6) FIG. 4B depicts an embodiment of an acoustic transducer component of the system.

(7) FIG. 5 depicts an embodiment of a component subsystem of the system.

(8) FIG. 6 depicts an embodiment of the submerged apparatus subsystem of the system.

(9) FIG. 7 depicts an embodiment of the underwater acoustic actuator-transponder unit of the system.

(10) FIG. 8 depicts a graphical representation of the geometry associated with the equations used in connection with an embodiment of the method of the present invention.

(11) FIG. 9 depicts an example of a display depicting submerged subsystem information.

### DETAILED DESCRIPTION OF THE INVENTION

(12) The present system and method is directed to the problem of ropeless fishing. Specifically, the present invention provides a system that replaces the functionality of traditional buoy and fixed line fishing systems and provides a method of use that allows for efficient and accurate communication between a vessel and submerged fishing gear, including the ability to identify the presence, ownership, location and association of said fishing gear, and the ability to cause said fishing gear to rise to the surface for retrieval.

(13) The present invention addresses the problems surrounding ropeless fishing. Lost fishing gear is costly both due to the replacement cost of the gear and the fact that lost gear necessarily reduces the amount of fishing gear that is yielding a catch to the fisherman. In addition, where a fisherman chooses to allocate time to attempt to find lost fishing gear, the time spent searching reduces the time available to pull and set additional gear, further reducing the fishing yield. The present invention addresses these issues by providing a system of ropeless fishing that enables communication between the user and submerged fishing gear, including the ability to identify the presence, ownership, location and association of said fishing gear, and the ability to cause said

fishing gear to rise to the surface for retrieval.

(14) Turning to FIG. 1, an embodiment of the ropeless fishing system **100** of the present invention is shown. The ropeless fishing system **100** comprises two main subsystems: a surface subsystem **10** and a submerged subsystem **60**. The ropeless fishing system **100** employs several methods to provide a complete implementation of a ropeless fishing system.

(15) Turning to FIG. 2, an embodiment of the surface subsystem **10** of the ropeless fishing system **100** is shown. The surface subsystem **10** is mounted on a vessel traversing the surface of a body of water. The surface subsystem **10** comprises an acoustic transceiver unit **20** and an acoustic transducer unit **40** and is capable of transmitting and receiving acoustic interrogation signals **41** to allow interrogation of multiple submerged subsystems **60** to determination the unique identification and location of each submerged subsystem **60** on the seafloor. In addition, the surface subsystem **10** is capable of commanding submerged subsystems **60** to rise to the surface for recovery.

(16) The acoustic transceiver unit **20** is comprised of a group of electronic components that control the operation of the ropeless fishing system **100** and provide interaction with the user. As depicted in FIG. 3, the electronic components of the acoustic transceiver unit **20** include a microcontroller **22**, which may be comprised of one or more printed circuit boards. The microcontroller **22** performs the processing functions of the surface subsystem **10**, including generating interrogation signals **41**, processing reply signals **61** to ascertain pertinent details of the responding submerged subsystems **60**, enabling the graphic representation of submerged subsystem **60** details on a human machine interface **25**, and processing input from a user in order to control the ropeless fishing system **100**. The microcontroller **22** is in electronic communication with a power amplifier **24** which amplifies the electronic signal received from the microcontroller **25** and passes it to a tuning component **26** prior to passing the tuned electronic signal to a transducer **44** of the acoustic transducer unit **40** for conversion of the electronic signal to an acoustic signal. In addition, the microcontroller **22** receives electronic signals from the acoustic transducer unit **40**, which may be processed to determine pertinent details regarding the submerged systems **60**, such as location and bearing.

(17) The microcontroller **22** is also in electronic communication with a human machine interface **25**. The human machine interface **25** enables interaction between the user and the ropeless fishing system **100**. Specifically, the human machine interface **25** includes a display capable of representing information regarding the submerged subsystems **60** of the ropeless fishing system **100**. For example, the display may be a chart plotter capable of depicting the location and bearing of submerged subsystems **60**. In addition to providing the ability to depict information regarding the submerged subsystems **60**, the human machine interface **25** may provide an interface capable of allowing a user to control the ropeless fishing system **100**. For example, the human machine interface **25** may be a touch-screen display capable of receiving user input to generate acoustic interrogation signals **41**.

(18) The human machine interface **25** may receive information from a GPS receiver **28** in order to determine information such as location and heading for the surface subsystem **10** and in order to calculate information regarding the submerged subsystems **60**. As described in more detail below, the surface subsystem **10** may be used to determine the depth of the seafloor. However, in some embodiments it may be advantageous to include a dedicated depth sounder **38** to provide water depth information. It is noted that while the components of the acoustic transceiver unit **20** are depicted in FIG. 3 as separate, these components may be physically grouped in any manner without deviating from the scope of the present invention. For example, the human machine interface **25** may be a tablet computer that includes a GPS receiver and includes a portion of the microcontroller processing capability.

(19) Turning the FIGS. 4A and 4B, two embodiments of the acoustic transducer unit **40** of the present invention are shown. The acoustic transducer unit **40** includes at least one transducer **42**, which converts the electronic signal passed from the acoustic transceiver unit **20** into an acoustic

signal that can be transmitted into the water. In embodiments where the acoustic transducer unit **40** comprises a single transducer **42**, the transducer **42** performs the transmit function with regard to outgoing acoustic interrogation signals **41** and also the receive function with regard to incoming acoustic reply signals **61**. When performing the receive function, the transducer **42** receives the acoustic reply signal **61** and converts the signal into an electronic signal that is passed back to the acoustic transceiver unit **20**. Alternatively, in some embodiments, the acoustic transducer unit **40** includes a first transducer **42**, which performs the receive function, and a second transducer **44**, which performs the transmit function. Regardless of the number of transducers utilized in the acoustic transducer unit **40**, the transmit function is omnidirectional in order to ensure that the acoustic interrogation signal **41** generated by the surface subsystem **10** is transmitted to all submerged systems **60** within range of the surface subsystem **10**, while the receive function is directional in order to facilitate calculation of submerged subsystem **60** information such as location and bearing.

(20) In order to facilitate the directional receive function of the transducer **42**, the acoustic transducer unit **40** is aligned such that the acoustic transducer unit **40** heading direction will coincide with the vessel heading direction. Further, because the transducers **42**, **44** must be located below the waterline in order to function, the acoustic transducer unit **40** may include a conduit **59** to create a waterproof environment for the cables used to connect the components of the acoustic transceiver unit **20** and the components of the acoustic transducer unit **40**. In addition, the acoustic transducer unit **40** may include a reply signal preamplifier **48** as shown in FIGS. **3** and **4A**. In this configuration, the reply signal preamplifier **48**, which is located within a housing **51**, receives the electronic signal resulting from the first transducer **42** and amplifies the signal before passing it on the acoustic transceiver unit **20**. Alternatively, a reply signal preamplifier **48** may be located elsewhere in the surface subsystem **10**. For example, as depicted in FIG. **4B**, the conduit may connect with a fitting **53** that enables a through-hull installation of the transducer **42** and the reply signal preamplifier **48** may be located within the hull of the vessel to provide additional protection for the electronics of the reply signal preamplifier **48**.

(21) Together, the acoustic transceiver unit **20** and the acoustic transducer unit **40** generate the outgoing acoustic interrogation signals **41** that will propagate from the surface vessel to multiple submerged subsystems **60** within range of the surface subsystem **10**. The outgoing acoustic interrogation signal **41** may be a simple acoustic signal. However, preferably, the acoustic interrogation signal **41** is a coded acoustic signal that enables the surface subsystem **10** to communicate multiple parameters within a single acoustic interrogation signal **41**. For example, a single acoustic interrogation signal **41** may include information concerning the identity of the surface subsystem **10** generating the acoustic interrogation signal **41** and information identifying the acoustic interrogation signal **41** as a request for a reply. The length and format of these coded acoustic interrogation signals **41** can be varied based on the size of the ropeless fishing system **100** and the desired level of security.

(22) Upon receiving the acoustic interrogation signal **41** transmitted by the acoustic transducer unit **40** each submerged subsystem **60** can generate an acoustic reply signal **61**. Similar to the acoustic interrogation signals **41**, the acoustic reply signals **61** are preferably coded acoustic signals. These acoustic reply signals **61** are received by the surface subsystem **10** and processed to compute the position of each submerged subsystem **60** relative to the transducer **42**. Where the acoustic reply signal **61** is a coded acoustic signal, the acoustic reply signal **61** can include information regarding the identification of the submerged system **60** and any association between multiple submerged subsystems **60** in order to allow the surface subsystem **10** to ascertain information regarding the submerged subsystem **60** such as ownership and the type of fishing gear deployed.

(23) Once an acoustic reply signal **61** is received by the acoustic transducer unit **40**, the acoustic signal is converted to an electrical signal and passed to the acoustic transceiver unit **20** for processing. The acoustic transceiver unit **20** computes the slant range by measuring the time of

transmission from the acoustic transducer unit **20** until the time of reception of the acoustic reply signal **61** transmitted by the submerged subsystem **60**. The outgoing acoustic interrogation signal **41** from the surface subsystem **10** to the submerged subsystem **60** can also be referred to as a downlink signal, while the acoustic reply signal **61** from the submerged subsystem **60** to the surface subsystem **10** can also be referred to as an uplink signal. Preferably, all of the acoustic signals utilized by the ropeless fishing system **100** are acoustic signals of short time duration and can be referred to as pings or pulses.

(24) In addition to utilizing the acoustic reply signal **61** to compute the slant range of the submerged subsystem **60**, the acoustic transceiver unit **20** can also use this signal to compute the relative bearing of the submerged subsystem **60**. For example, where the acoustic transducer unit **40** is installed on the vessel such that the acoustic transducer unit **40** heading direction coincides with the surface vessel heading direction, the relative bearing is the angle from the ships heading to the direction of arrival of the acoustic reply signal **61**.

(25) As described above, the acoustic transceiver unit **20** is capable of receiving vessel location and heading information from a GPS receiver **28**, which may be physically grouped with other components of the acoustic transceiver unit **20** or may be a separate, external GPS receiver **28**. The acoustic transceiver unit **20** is further capable of receiving water depth information from a dedicated depth sounder **38** that is installed and configured to communicate its depth information to the acoustic transceiver unit **20**. Alternatively, the acoustic transceiver unit **20** can directly calculate water depth by calculating the time delay of acoustic interrogation pulses **41** that have been reflected back to the acoustic transducer unit **40** as echoes off of the seafloor. In addition, where the ropeless fishing system **100** utilizes coded acoustic signals that include identification and association information, the acoustic transceiver unit **20** will be able to determine the identification, location and association of each submerged subsystem **60** with a single acoustic interrogation signal **41** and a single acoustic reply signal **61**. Once these characteristics are identified, the acoustic transceiver unit **20** can determine not only location and bearing, but also the ownership of various submerged subsystems **60** as well as the nature of the fishing gear deployed on the seafloor, e.g., trawls or individual traps.

(26) The acoustic transceiver unit **20** is capable of computing the geodetic location (lat, lon) of each submerged subsystem **60** using the slant range, relative bearing, water depth, vessel position and vessel heading. Once computed, the acoustic transceiver unit **20** will transmit the submerged subsystem **60** information, including geodetic location, depth, apparatus type, and apparatus ownership, to the human machine interface **25**. Preferably, this information is then presented on a display so the user can make informed decisions regarding gear deployment and retrieval.

(27) Turning to FIGS. 5-7, exemplar embodiments of the submerged subsystem **60** and the underwater acoustic actuator-transponder unit (UAAT) **62** of the ropeless fishing system **100** of the present invention are depicted. The submerged subsystem is comprised of three components: a UAAT **62**, a compressed gas source **82**, and an enclosed flotation bag **84**.

(28) The UAAT **62** performs the function of communicating with the surface subsystem **10** by receiving the acoustic interrogation signal **41** and transmitting the acoustic reply signal **61**. In addition, the UAAT **62** also controls the release of gas from the compressed gas source **82** in order to inflate the enclosed flotation bag **84**. The UAAT **62** includes a housing **66** and a transducer portion **70**. The transducer portion **70** includes at least one transducer **71**. Similar to the acoustic transducer unit **40** described above, the UAAT **62** may utilize a single transducer **71** that performs the receive function with regard to incoming interrogation signals **41** and the transmit function with regard to outgoing reply signals **61**. Alternatively, the UAAT **62** may include a first transducer **71**, which performs the receive function, and a second transducer **72**, which performs the transmit function.

(29) The housing **66** is waterproof and submersible in order to protect the electronic components that comprise an acoustic signal processor and valve control unit (ASP-VCU) **68** and an energy



source **67** such as a battery. In addition, the housing may include an LED **63**, which provides the ability to signal the status of the UAAT to the user without requiring the user to open the housing **66**, a vacuum port **64**, which enables air to be removed from the housing **66** in order to create a vacuum, and a multi-control port **65** which enables communication with the ASP-VCU **68** and the power source **68**.

(30) The transducer portion **70** is connected to the housing **66** in a manner that maintains the waterproof and submersible nature of the housing **66** and enables the transducers **71**, **72** to remain in electronic communication with the ASP-VCU **68**. While the transducers **71**, **72** are depicted in a transducer portion **70** that is directly connected to the housing **66**, the transducers may also be positioned away from the housing provided the transducers **71**, **72** remain in electronic communication with the ASP-VCU **68**.

(31) The ASP-VCU **68** is comprised of a microcontroller and one or more electronic components that process the electronic signals received from and transmitted to the transducers **71**, **72**, control the power supply, and control the operation of a solenoid valve **80**. When an acoustic interrogation signal **41** is received by the transducer **71**, the acoustic signal is converted to an electronic signal by the transducer **71** and then passed to the ASP-VCU **68** for processing. The processing performed by the ASP-VCU can also include amplification and interpretation of the electronic signal. In addition, depending on the nature of the acoustic interrogation signal **41**, the ASP-VCU **68** can either generate a reply signal or operate a solenoid valve **80**. As described above with regard to interrogation signals, the reply signal generated by the ASP-VCU **68** can be a coded signal and can include information regarding the identification and association of the submerged subsystem **60**.

(32) The multi-control port **65** enables communication with the ASP-VCU **68** and the power source **67**. As depicted in FIGS. 5 and 6, a control cable **78** is connected to the multi-control port **65** via a connector **75**. In this configuration, the connector **75** creates an electronic connection between the ASP-VCU **68** and the power source **67** inside the housing **66** and the solenoid valve **80**, such that the ASP-VCU **68** can control the operation of the solenoid valve **80**. In addition, the connector **75** can function as an activation switch for the UAAT **62** by creating the necessary electric connection between the power source **67** and the ASP-VCU **68** once the connector **75** is secured to the multi-control port **65**. In alternative configurations, the multi-control port **65** can be connected to a separate cable for the purposes of testing and reprogramming the ASP-VCU **68** and charging the battery **67**.

(33) In some situations, it is advantageous to create a vacuum inside the chamber in order to minimize the possibility of condensation inside the housing **66**. Accordingly, a vacuum port **64** may be included on the housing. In addition to minimizing the risk of condensation, creating a vacuum inside the housing **66** can also be used to test the integrity of the housing **66**. For example, a sensor can be included within the ASP-VCU **68** to detect when the housing **66** has lost vacuum, which can be used to activate the LED **63** to indicate a system error.

(34) The UAAT **62** is connected to the solenoid valve **80** via the control cable **78**. The solenoid valve is also connected to the compressed gas cylinder **82** and the enclosed floatation bag **84** such that the solenoid valve **80** controls the flow of gas between the compressed gas cylinder **82** and the enclosed floatation bag **84**. When the UAAT **62** receives an acoustic interrogation signal **41** designed to cause the submerged subsystem **60** to rise to the surface of the water, a signal is generated by the ASP-VCU **68** and transmitted from the UAAT **62** to the solenoid valve **80** via the control cable **78**. The signal causes the actuation of the solenoid valve **80** and permits gas to flow from the compressed gas cylinder **82** into a gas line **88** connecting the solenoid valve **80** and the enclosed floatation bag **84**. As the compressed gas fills the enclosed floatation bag **84**, the submerged subsystem **60** is lifted from the seafloor and floats to the surface for recovery.

(35) Although the solenoid valve **80** is depicted as being external to the housing **66** of the UAAT **62**, alternatively, the solenoid valve **80** may be located within the housing **66**. In configurations where the solenoid valve **80** is located within the housing **66**, the housing **66** may further include

inlet and outlet ports for the gas line **88**. In this configuration, the gas line **88** may be connected to the inlet and outlet ports such that the compressed gas source **82** and the enclosed floatation bag **84** are placed in fluid communication via the solenoid valve **80** located within the housing.

(36) The compressed gas source **82** may be any appropriate pressure vessel containing compressed gas. Further, the compressed gas may be any compressed gas known in the art, including air, carbon dioxide, nitrogen, oxygen, or helium. The compressed gas source **82** may or may not include a manually operated valve. For example, the compressed gas source may be a carbon dioxide cartridge that may get pierced upon connection to the solenoid valve **80**.

(37) The enclosed floatation bag **84**, which is also known in the art as a lift bag or a balloon, may include a pressure relief valve. The pressure relief valve can help prevent rupture, tearing, or other damage that would cause gas leaks upon complete inflation or gas expansion due to reduced hydrostatic pressure during ascent, limiting the risk that damage resulting from overinflation would cause release of gas from the enclosed floatation bag **84** and reduce buoyancy and lift capacity. The enclosed floatation bag **84** has an inlet port **86** to accept compressed gas into the bladder to allow expansion and generation of lifting (buoyancy) forces.

(38) The volume of the enclosed floatation bag **84** and the compressed gas source **82** can be varied to create sufficient lifting forces to bring the submerged subsystem to the surface for retrieval. For example, a typical 19 cu. ft. compressed gas SCUBA cylinder would provide enough air to fully inflate a 500 pound enclosed floatation bag **84** at a depth of 80 feet of seawater, while a standard 80 cu. ft. compressed gas SCUBA cylinder would provide the ability to inflate the same size 500 pound enclosed floatation bag **84** to a depth of 338 feet of seawater.

(39) As shown in FIG. 5, the submerged subsystem **60** is attached to a piece of fishing gear, such as a trap. In order to protect the components of the submerged subsystem **60**, the submerged subsystem **60** may be attached to and surrounded by a frame **95**, with the submerged subsystem attached to the fishing gear **92** via the frame **95**. In addition, where multiple pieces of fishing gear **92** are deployed as a trawl **90**, not all of the fishing gear **92** may include a submerged subsystem **60**. Preferably, fishing gear **92** deployed as a trawl **90** will include a submerged subsystem **60** at each end of the trawl **90** as depicted in FIG. 1. In this configuration, it is also preferable that both of the submerged subsystems **60** utilized in the trawl **90** include an association indicator in their acoustic reply code **61**, which will allow the surface subsystem **10** to identify that the two submerged subsystems **60** represent the ends of a trawl **90**.

(40) Turning to a method of using the ropeless fishing system **100** to identify the position of submerged subsystems **60**, a surface subsystem **10** comprises an acoustic transceiver unit **20** and an acoustic transducer unit **40** located on a vessel operating at the surface of the water. As previously described, the acoustic transceiver unit **20** is connected to the acoustic transducer unit **40**. The acoustic transceiver unit **20** generates an electronic interrogation signal that is transmitted through a power amplifier **24** and a tuning component **26** to a transducer **44**, where the electronic interrogation signal is converted to an acoustic interrogation signal **41** that is transmitted into the ocean as an omnidirectional acoustic signal. This acoustic interrogation signal **41** is reflected back from the seafloor as an echo and can be processed by the surface subsystem **10** to obtain depth in a manner similar to a traditional echosounder or fathometer. In addition, the acoustic interrogation signal **41** is also received by the UAAT **62** of each submerged subsystem **60** within range of the surface subsystem **10**, which will result in the UAAT **62** generating and transmitting an acoustic reply signal **61** that is sent back to the surface apparatus **10**. Acoustic reply signals **61** are by necessity different than the acoustic interrogation signals **41**. However, acoustic reply signals **61** may also be different between each submerged subsystem **60**. The echoes of the acoustic interrogation signal **41** and the acoustic reply signals **61** are received by the surface subsystem **10**, where the acoustic signals are converted to electronic signals by the transducer **42** and transmitted to the acoustic transceiver unit **20** for processing.

(41) Using the speed of sound in water, the time of the acoustic reply signals **61** can be converted

to the geometric slant range between the acoustic transducer unit **40** and the submerged subsystem **60**. The acoustic transducer unit **40** receives the acoustic reply signal **61** by means of a receiving hydrophone. The receiving hydrophone is directional such that it can allow measurement of the angle of arrival in azimuth of the signal over the full 360 degrees around the transducer **42**. By installing the acoustic transducer unit **20** such that the acoustic transducer unit **40** heading direction will coincide with the vessel heading direction, the measurement of the angle of arrival produces a measurement of relative bearing of the submerged subsystem **60** from the surface vessel.

(42) Once the acoustic transceiver unit **20** has calculated the slant range (R), relative bearing ( $\theta_{\text{sub.R}}$ ), heading (H) and depth (d), the position of the submerged subsystem **60** can be computed as follows: 1. Compute the horizontal distance (h) between the submerged subsystem **60** and acoustic transducer unit **40** according to:

$h = \sqrt{R^2 - d^2}$  2. Compute the true bearing ( $\theta_{\text{sub.T}}$ ) from the relative bearing and heading. 3. Compute the Latitude ( $\phi$ ) and Longitude ( $\lambda$ ) according to:

$$(43) \quad \begin{aligned} \phi_B &= \phi_S + \frac{h \sin \theta_{\text{sub.T}}}{R_E} & \lambda_B &= \lambda_S + \frac{h \cos \theta_{\text{sub.T}}}{R_E} \end{aligned}$$

Where ( $\phi_{\text{sub.B}}$ ,  $\lambda_{\text{sub.B}}$ ) is the latitude and longitude of the submerged subsystem **60**, and ( $\phi_{\text{sub.S}}$ ,  $\lambda_{\text{sub.S}}$ ) is the latitude and longitude of the acoustic transducer unit **40**. Note that this is a simplified computation sufficiently accurate over short distances and for this application. We could alternatively implement a solution using the Vincenty algorithm. FIG. **8** provides a graphical depiction of the geometry associated with the equations written in this section.

(44) After computing the latitude and longitude of a submerged subsystem **60**, the acoustic transceiver unit **20** can display the information via the human machine interface **25**. For example, the information may be transmitted in NMEA-2000 or NMEA-0183 format and presented on a display as shown in FIG. **9**. Furthermore, the information could be sent as a Waypoint or as an AIS message. Alternatively, the information can include the identification and association information transmitted in the acoustic interrogation signal **41** and the acoustic reply signal **61** to display submerged fishing gear as individual traps and trawls, with relevant ownership information, on properly modified chart plotting software.

(45) Utilizing coded acoustic reply signals **61** provides the ability to differentiate between different submerged subsystems **60** in situations where a plurality of submerged subsystems **60** are deployed in a given area of the seafloor. This differentiation can be used to identify which submerged subsystems **60** are owned by the operator or can be used by regulators and law enforcement to identify the ownership of submerged subsystems deployed on the seafloor.

(46) Where fishing gear **92** is deployed on the seafloor as a trawl **90**, the trawl **90** may utilize a submerged subsystem **60** at each end of the trawl **90**. Each of these submerged subsystems **60** may be coded to transmit an acoustic reply signal **61** that allows the surface subsystem **10** to identify the submerged subsystems **60** as connected and the appropriate message can be displayed on the human machine interface **25** to represent the location of submerged subsystems **60** with a string of fishing gear **92** connected between them.

(47) Turning to FIG. **9**, an example of a display portion of the human machine interface **25** depicting location, ownership and association information of a plurality of submerged subsystems **60** is shown. In the example, the location information determined by the surface subsystem **10** is used to plot each submerged subsystem **60** within range of the surface subsystem **10**. In addition, identification information provided by each submerged subsystem **60** is represented as an “o” where the submerged subsystem **60** is owned by the user, while each submerged subsystem **60** is represented by an “x” where the submerged subsystem is owned by someone other than the owner. Finally, association information is represented by the dashed line connecting submerged subsystem **60**, which indicates that the submerged subsystems **60** are connected to form a trawl **90**. It will be understood that this example represents only one means of displaying the information ascertained by the surface subsystem **10**, and that different symbols, colors or other indicators could be used to

similar effect.

(48) Turning to a second method of the present invention, the surface subsystem **10** can be used to signal the submerged subsystem **60** to inflate the enclosed floatation bag **84**, causing the submerged subsystem **60** to float to the surface for recovery. A first step includes utilizing an acoustic interrogation signal **41** to compute the location of one or more submerged subsystems **60** as described above. Once the one or more submerged subsystems **60** are located, a second acoustic interrogation signal **61** can be transmitted by the acoustic transponder unit **40**. This second acoustic interrogation signal **61** can be coded such that it includes a code designed to instruct the submerged subsystem **60** to inflate the enclosed floatation bag **84**. Upon receipt of this second coded interrogation signal **61** by each submerged subsystem **60**, the solenoid valve **80** is actuated, allowing gas from the compressed gas source **82** to the enclosed floatation bag **84**. As the gas fills the enclosed floatation bag **84**, the submerged subsystem **60** is lifted from the seafloor and floats to the surface for retrieval.

(49) In some situations, it is advantageous to embed additional information in the coded acoustic interrogation signal **41** in order to restrict the communication of the acoustic interrogation signal **41** to specific submerged subsystems **60** or a specific group of submerged subsystems **60**. Therefore, the second acoustic interrogation signal **41** may be coded such that the acoustic interrogation signal **41** is targeted to a specific submerged subsystem **60** and includes a code designed to instruct the targeted submerged subsystem **60** to inflate its enclosed floatation bag **84**. Further, where a user desires to instruct multiple submerged subsystems **60** to rise to the surface, multiple second interrogation signals **41** may be transmitted by the acoustic transponder unit **40**, with each acoustic interrogation signal **41** coded to instruct an individual submerged subsystem **60**.

(50) Where a targeted acoustic interrogation signal **41** is received by the appropriate submerged subsystem **60**, the submerged subsystem **60** will trigger the actuation of the solenoid valve **80** and the submerged subsystem **60** will float to the surface. However, where the targeted acoustic interrogation signal **41** is received by a submerged system **60** that is not a target submerged subsystem **60**, the acoustic interrogation signal **41** will not result in the actuation of the solenoid valve and the non-targeted submerged subsystem **60** will remain on the seafloor.

(51) It will be understood that the methods described herein enable a user to generate a single acoustic interrogation signal **41**, which is broadcast in an omnidirectional pattern from the surface subsystem **10** and receive multiple acoustic reply signals **61**, each reply signal representing an individual submerged subsystem **60** within range of the surface subsystem **10**. Accordingly, location, identification, and association information can be ascertained and/or calculated for each of the acoustic reply signals **61** received by the surface subsystem **10**. In practice, this efficient communication method, which obtains information regarding multiple submerged subsystems with a single outbound acoustic signal, precludes the need for the vessel to maneuver and minimizes the total number of acoustic transmissions present in the water.

(52) It is noted that the foregoing examples have been provided merely for the purpose of explanation and are in no way to be construed as limiting of the present invention. While the present invention has been described with reference to exemplary embodiments, it is understood that the words, which have been used herein, are words of description and illustration, rather than words of limitation. Changes may be made, within the purview of the appended claims, as presently stated and as amended, without departing from the scope and spirit of the present invention in its aspects. Although the present invention has been described herein with reference to particular means, materials and embodiments, the present invention is not intended to be limited to the particulars disclosed herein; rather, the present invention extends to all functionally equivalent structures, methods and uses, such as are within the scope of the appended claims.

## Claims

1. A ropeless fishing system comprising: a. a surface subsystem installed on a vessel, the surface subsystem comprising an acoustic transceiver unit in communication with an acoustic transducer unit, the surface subsystem configured to generate and omnidirectionally transmit a first acoustic interrogation signal into a body of water surrounding the vessel and further configured to directionally receive acoustic reply signals; and b. at least one submerged subsystem submerged in the body of water, each of the at least one submerged subsystem comprising a compressed gas source and an enclosed flotation bag in fluid communication and an underwater acoustic actuator-transponder unit configured to receive the acoustic interrogation signal transmitted by the surface subsystem and to transmit an acoustic reply signal into the body of water and further configured to control the flow of gas between the compressed gas source and the enclosed flotation bag; c. the surface subsystem further configured to process each acoustic reply signal by the or each submerged subsystem and received by the surface subsystem to determine the location, ownership, and association of the or each submerged subsystem and to generate and transmit a second acoustic interrogation signal, the second acoustic interrogation signal causing the or at least one submerged subsystem to rise to the surface of the body of water.
2. The ropeless fishing system of claim 1 wherein the acoustic transducer unit comprises a first transducer configured to directionally receive the acoustic reply signal and a second transducer configured to omnidirectionally transmit the acoustic interrogation signal.
3. The ropeless fishing system of claim 1 wherein the at least one submerged subsystem is a plurality of submerged subsystems.
4. The ropeless fishing system of claim 1 wherein the acoustic interrogation signal and the acoustic reply signal are coded acoustic signals that include information regarding identification and any association between multiple submerged subsystems.
5. The ropeless fishing system of claim 1 wherein the location, ownership and association of each of the at least one submerged subsystem is displayed to a user on a human machine interface.
6. The ropeless fishing system of claim 1 wherein the second acoustic interrogation signal is targeted to a specific submerged subsystem.
7. A method of ropeless fishing comprising: a. providing a ropeless fishing system comprising: i. a surface subsystem installed on a vessel, the surface subsystem comprising an acoustic transceiver unit in communication with an acoustic transducer unit, the surface subsystem configured to omnidirectionally transmit a first acoustic interrogation signal into a body of water surrounding the vessel and further configured to directionally receive acoustic reply signals; and ii. at least one submerged subsystem submerged in the body of water, each of the at least one submerged subsystem comprising an underwater acoustic actuator-transponder unit configured to receive the acoustic signal transmitted by the surface subsystem and to transmit an acoustic signal into the body of water and further configured to control the flow of gas between a compressed gas source and an enclosed flotation bag; iii. the surface subsystem further configured to process each acoustic reply signal by the or each submerged subsystem and received by the surface subsystem to determine the location of the or each submerged subsystem and to generate and transmit a second acoustic interrogation signal, the second acoustic interrogation signal causing the or at least one submerged subsystem to rise to the surface of the body of water; b. generating and omnidirectionally transmitting an acoustic interrogation signal with the surface subsystem; c. receiving the acoustic interrogation signal with one of the at least one submerged subsystem; d. processing the received acoustic interrogation signal with the underwater acoustic actuator-transponder unit and generating and transmitting an acoustic reply signal; e. receiving the acoustic reply signal with the surface subsystem; f. measuring a relative bearing of the acoustic reply signal and calculating slant range and true bearing; g. calculating the location of the submerged subsystem; h. generating and transmitting a second acoustic interrogation signal with the surface subsystem; i. receiving the second acoustic interrogation signal with one of the at least one submerged

subsystem; j. processing the received second acoustic interrogation signal with the underwater acoustic actuator-transponder unit and generating and transmitting a control signal that actuates a solenoid valve, causing gas from the compressed gas source to flow into and inflate the enclosed floatation bag and causing the one of the at least one submerged subsystem to rise to the surface of the body of water.

8. The method of ropeless fishing of claim 7 wherein the surface subsystem further comprises a human machine interface having a display and further comprising displaying the location of the submerged subsystem to a user on the human machine interface.

9. The method of ropeless fishing of claim 7 wherein the at least one submerged subsystem is a plurality of submerged subsystems and wherein the acoustic interrogation signal is received by a plurality of the plurality of submerged subsystems.

10. The method of ropeless fishing of claim 9 wherein steps e through g are repeated for each acoustic reply signal received from each of the plurality of submerged subsystems.

11. The method of claim 10 further comprising the step of displaying the location of each of the plurality of submerged subsystems to a user on the human machine interface.

12. The method of ropeless fishing of claim 9 further comprising retrieving the one of the at least one submerged subsystem once the submerged subsystem rises to the surface of the body of water.

13. A method of ropeless fishing comprising: a. providing a ropeless fishing system comprising: i. a surface subsystem installed on a vessel, the surface subsystem comprising an acoustic transceiver unit in communication with an acoustic transducer unit, the surface subsystem configured to omnidirectionally transmit a first acoustic interrogation signal into a body of water surrounding the vessel and further configured to directionally receive acoustic reply signals; and ii. at least one submerged subsystem submerged in the body of water, each of the at least one submerged subsystem comprising an underwater acoustic actuator-transponder unit configured to receive the acoustic signal transmitted by the surface subsystem and to transmit an acoustic signal into the body of water; and iii. the surface subsystem further configured to process each acoustic reply signal by the or each submerged subsystem and received by the surface subsystem to determine the location of the or each submerged subsystem and to generate and transmit a second acoustic interrogation signal, the second acoustic interrogation signal causing the or at least one submerged subsystem to rise to the surface of the body of water; b. generating and omnidirectionally transmitting an acoustic interrogation signal with the surface subsystem; c. receiving the acoustic interrogation signal with one of the at least one submerged subsystem; d. processing the received acoustic interrogation signal with the underwater acoustic actuator-transponder unit and generating and transmitting an acoustic reply signal; e. receiving the acoustic reply signal with the surface subsystem; f. measuring a relative bearing of the acoustic reply signal and calculating slant range and true bearing; g. calculating the location of the submerged subsystem.

14. The method of ropeless fishing of claim 13 wherein the surface subsystem further comprises a human machine interface having a display and further comprising displaying the location of the submerged subsystem to a user on the human machine interface.

15. The method of ropeless fishing of claim 13 wherein the at least one submerged subsystem is a plurality of submerged subsystems and wherein the acoustic interrogation signal is received by a plurality of the plurality of submerged subsystems.

16. The method of ropeless fishing of claim 15 wherein steps e through g are repeated for each acoustic reply signal received from each of the plurality of submerged subsystems.

17. The method of claim 16 further comprising the step of displaying the location of each of the plurality of submerged subsystems to a user on the human machine interface.

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